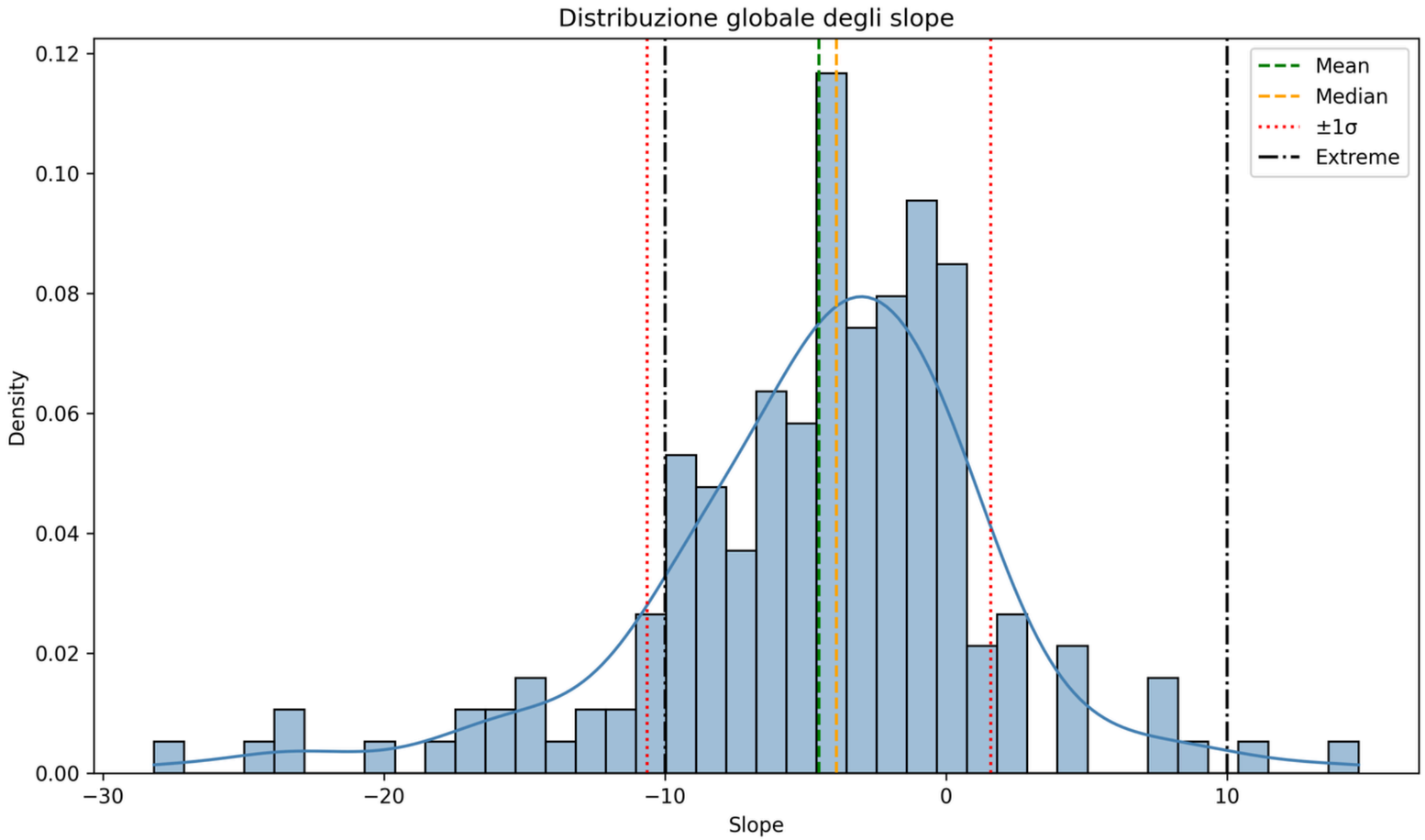
PROGRESS PREDICTION

Global distribution of progression slopes

Mean : -4.5
Median : -3.9

78% of samples lie within one standard deviation from the mean.
Extreme cases exist on both tails, but they represent a relatively small fraction of the dataset.



Strong bias toward mean predictions, increasing the risk of mode collapse during model training.

Fold distribution of progression slopes

The concentration of targets around the mean is consistent across all folds and splits, indicating a dataset-level imbalance rather than a splitting artifact.

Fold	Mean slope	Std	Extreme %	±1σ concentrion
0	-4.56	5.97	13.5%	76%
1	-4.74	5.84	13%	79.8%
2	-4.16	5.86	13.6%	77.2%
3	-4.33	5.72	13%	74.3%
4	-4.07	5.99	13%	77.5%

Across all folds:
75-80% of samples lie within ±1σ ,the distribution is strongly peaked.
Extremes are rare and diluted

Consequences in training the CNN for slopes:

- Loss dominated by central mass
- Gradients from extremes are rare
- Oversampling helps exposure

Tried with:

- Sample weighting
- Oversampling extremes

But model still collapses

Example training with Oversampling the CNN:

FVC MAE : 237.20 ± 29.80 ml
Accuracy: 0.418
Precision : 0.519
Recall: 0.094
Specificity: 0.868
AUC: 0.549

	Stable (Predicted)	Progression (Predicted)
Stable (True)	12.0	1.8
Progression (True)	17.4	1.8

On average the CNN predicts progression for only 1.8 patients per fold , while missing more than 17 progression cases per fold.

This due to the model collapse

Therefore it's probably certain (need to do experiment) that:

- Directly predict FVC at week 52
- Direct Progression prediction

Will be better!

Possible pivots for the slope model:

- Multi-task learning:
  - two complementary objectives
    - Head 1: Regression task → slope
    - Head 2: Classification task → predict probability of progression

We have therefore two losses :

- Regression loss → push all samples to the middle
- Classification loss → pushes progressors and non progressors apart, penalizing patients being all stable

- Heteroscedastic regression (predicting uncertainty) → addresses label noise

Instead of predicting one single value for the slope , the network predict:

slope - Normal(μ, σ^2)

Where:

- μ = predicted slope
- σ = predicted uncertainty (per patient)

With loss function (Gaussian NLL) :

$$\mathcal{L} = \frac{(y-\mu)^2}{\sigma^2} + \log(\sigma^2)$$

- If the model is unsure → it can increase σ
- But increasing σ is penalized by $\log(\sigma^2)$
- So the model must justify uncertainty with real errors

With NLL predicting the mean with low σ is heavily penalized.